COMSATS UNIVERSITY ISLAMABAD, ATTOCK CAMPUS

Department of Computer Sciences

Course Name: Information Security Lab

Assignment # 04

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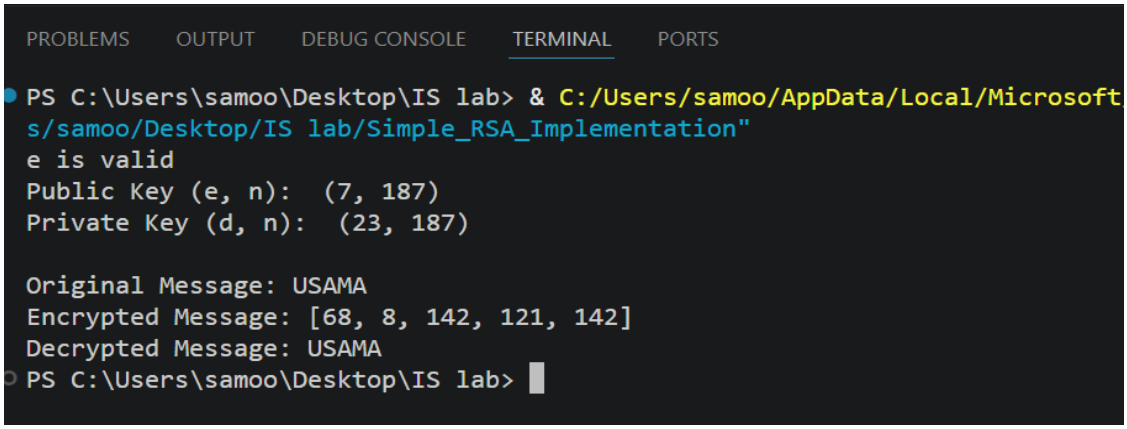
Task 1: Simple RSA Implementation (Without Libraries)

Code:

```
# Task 1: Simple RSA Implementation Without Libraries

# Function to calculate Greatest Common Divisor (GCD)
def gcd(a, b):
    while b != 0:
        a, b = b, a % b
    return a


# Function to calculate modular inverse
def mod_inverse(e, phi):
    for d in range(1, phi):
        if (d * e) % phi == 1:
            return d
    return None


# Function for RSA Encryption
def encrypt(message, e, n):
    encrypted = []

    for char in message:
        # Convert character into ASCII value
        m = ord(char)

        # RSA Encryption Formula
        c = (m ** e) % n

        encrypted.append(c)

    return encrypted

# Function for RSA Decryption
def decrypt(ciphertext, d, n):
    decrypted = ""

    for c in ciphertext:
        # RSA Decryption Formula
        m = (c ** d) % n

        # Convert ASCII back to character
```

```
        decrypted += chr(m)

    return decrypted
# ---------------- RSA Key Generation ----------------

p = 17
q = 11

# Step 2: Calculate n
n = p * q

# Step 3: Calculate Euler Totient Function
phi = (p - 1) * (q - 1)

# Step 4: Choose e
e = 7

# Check if e and phi are coprime
if gcd(e, phi) == 1:
    print("e is valid")
else:
    print("Choose another e")

# Step 5: Calculate d
d = mod_inverse(e, phi)

# Display Keys
print("Public Key (e, n): ", (e, n))
print("Private Key (d, n): ", (d, n))

# ---------------- Message Encryption ----------------

message = "USAMA"

print("\nOriginal Message:", message)

encrypted_message = encrypt(message, e, n)

print("Encrypted Message:", encrypted_message)

# ---------------- Message Decryption ----------------

decrypted_message = decrypt(encrypted_message, d, n)

print("Decrypted Message:", decrypted_message)
```

Output:



Line By Line Explanation:

gcd() Function

def gcd(a, b):

This function is created to calculate the Greatest Common Divisor.

while b != 0:

Loop runs until b becomes 0.

a, b = b, a % b

Euclidean Algorithm is used.

return a

Returns final GCD value.

mod_inverse() Function

def mod_inverse(e, phi):

This function calculates private key d.

for d in range(1, phi):

Checks every value from 1 to phi.

if (d * e) % phi == 1:

Condition for modular inverse.

return d

Returns the correct value of d.

Prime Numbers

p = 11

q = 13

Two prime numbers are selected.

Calculate n

n = p * q

Multiplies both prime numbers.

Result:

143

Calculate phi

phi = (p - 1) * (q - 1)

Calculates Euler Totient Function.

Result:

120

Public Key e

e = 7

Public exponent is selected.

Verify e

if gcd(e, phi) != 1:

Checks whether e and phi are coprime.

Private Key d

d = mod_inverse(e, phi)

Calculates modular inverse.

Result:

103

Message

```
message = "USAMA"
```

Text message that will be encrypted.

ASCII Conversion

```
message_numbers = [ord(char) for char in message]
```

Converts characters into ASCII values.

Example:

U = 85
S = 83

Encryption

```
cipher = (num ** e) % n
```

RSA encryption formula is applied.

Formula:

C = M^e mod n

Decryption

```
plain = (cipher ** d) % n
```

RSA decryption formula is applied.

Formula:

M = C^d mod n

Convert Back to String

```
final_message = ''.join(decrypted)
```

Combines characters into original message.

Advantages of RSA

1. RSA provides secure communication.
2. Public and private keys are different.
3. Private key is never shared.
4. RSA is widely used in HTTPS and secure systems.

Task 2 – Digital Signature Using RSA

Code:

```
# Task 2: Digital Signature Using RSA

import hashlib
# Function to calculate GCD

def gcd(a, b):
    while b != 0:
        a, b = b, a % b
    return a


# Function to calculate modular inverse

def mod_inverse(e, phi):
    for d in range(1, phi):
        if (d * e) % phi == 1:
            return d


# RSA Key Generation
p = 17
q = 11

n = p * q
phi = (p - 1) * (q - 1)

e = 7

if gcd(e, phi) != 1:
    print("Invalid e")
```

```
d = mod_inverse(e, phi)


# Public and Private Keys
public_key = (e, n)
private_key = (d, n)

print("Public Key:", public_key)
print("Private Key:", private_key)


# Original Message
message = "Information Security Lab"


# Step 1: Create SHA256 Hash
hash_value = hashlib.sha256(message.encode()).hexdigest()

print("Original Hash:", hash_value)


# Convert hash into integer
hash_int = int(hash_value, 16)


# Step 2: Create Digital Signature
# Signature = hash^d mod n

signature = pow(hash_int, d, n)

print("Digital Signature:", signature)
```

Output:

```
PS C:\Users\samoo\Desktop\IS lab> & C:/Users/samoo/AppData/Local/Microsoft/WindowsApps/python3.1
s/samoo/Desktop/IS lab/Digital_Signature"
Public Key: (7, 187)
Private Key: (23, 187)
Original Hash: 490572601061b11440ac11cff7a7529b4cd153837b66f09216346b4e9b403a69
Digital Signature: 70
PS C:\Users\samoo\Desktop\IS lab>
```

Line by Line Explanation:

Import hashlib

```
import hashlib
```

This library is used to create SHA256 hash.

RSA Key Generation

```
p = 17
```

```
q = 11
```

Two prime numbers are selected.

```
n = p * q
```

Calculates n.

Result:

187

```
phi = (p - 1) * (q - 1)
```

Calculates Euler Totient Function.

Result:

160

```
e = 7
```

Public exponent.

```
d = mod_inverse(e, phi)
```

Private key is calculated.

Result:

23

Message Hashing

```
hash_value = hashlib.sha256(message.encode()).hexdigest()
```

Creates SHA256 hash of message.

Hash functions convert data into fixed-size output.

Convert Hash to Integer

```
hash_int = int(hash_value, 16)
```

Converts hexadecimal hash into integer.

Signature Creation

```
signature = pow(hash_int, d, n)
```

Creates digital signature using private key.

Formula:

Signature = Hash^d mod n

Signature Verification

```
verified_hash = pow(signature, e, n)
```

Decrypts signature using public key.

Formula:

Hash = Signature^e mod n

Compare Hashes

```
if verified_hash == original_hash_check:
```

Checks whether both hashes match.

If matched:

Message is authentic.

Modified Message Test

```
modified_message = "Information Security lab"
```

Message is slightly modified.

Only one letter is changed.

Verification Failure

```
if verified_hash == (modified_hash_int % n):
```

Hashes become different.

Verification fails.

This proves message integrity.

Advantages of Digital Signature

1. Confirms sender identity.
2. Prevents message tampering.
3. Provides data integrity.
4. Prevents denial by sender.

Why Verification Fails After Modification

Even a small change in message changes SHA256 hash completely.

Example:

Original:

Information Security Lab

Modified:

Information Security lab

Only one letter changes, but hash becomes completely different.

This property is called:

Avalanche Effect

Real World Applications

Digital signatures are used in:

- Banking systems
- Software verification
- E-commerce
- Secure emails
- Blockchain technology